

General Finite-Exponent Matrix Hypercontractivity over Large Alphabets

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Source Question

Arunachalam and Doriguello ask in Future work item 4 of arXiv:2109.02600 whether their matrix-valued $(2, p)$ hypercontractive theorem can be upgraded to a general (q, p) hypercontractive statement, first over the Boolean cube and then over $\mathbb{Z}_r^n$. The source passage is shown in Figure 1.

3. Better bounds on streaming algorithms. What is the quantum space complexity of approximating Unique Games? Given the recent work of Kallaugher and Parekh [45], who proved a $\Omega(n)$ streaming lower bound for Max- t -Cut, we conjecture that the same bound applies to Unique Games. A proof might require obtaining new quantum lower bounds for the communication problems introduced in [19, 20, 47, 48].

4. Generalized hypercontractivity. Another open question is regarding our main Result 1, which shows a form of $(2, p)$ -hypercontractivity, since the result works for all Schatten p -norms with $p \in [1, 2]$. Can we prove a general (q, p) -hypercontractive statement for matrices, firstly for matrix-valued functions over $\{0, 1\}$, and then further generalize that to functions over $\mathbb{Z}_r$? Proving this might also require a different generalization of the inequality of Ball, Carlen, and Lieb [5].

2 Preliminaries

Let $[n] \triangleq \{1, \dots, n\}$. For $r \in \mathbb{Z}$, $r \geq 2$, we let $\mathbb{Z}_r \triangleq \{0, \dots, r-1\}$ be the ring with addition and multiplication modulo r , and let $\omega_r \triangleq e^{2\pi i/r}$. Given $S \in \mathbb{Z}_r^n$, we write $|S| \triangleq |\{i \in [n] : S_i \neq 0\}|$ for its Hamming weight. Let $D(\mathbb{C}^m)$ be the set of all quantum states over $\mathbb{C}^m$, i.e., the set of positive semi-definite matrices with trace 1. For a matrix $M \in \mathbb{C}^{m \times m}$, its (unnormalized) Schatten p -norm is defined as $\|M\|_p \triangleq (\text{Tr } |M|^p)^{1/p} = (\sum_{i=1}^m \sigma_i(M)^p)^{1/p}$, where $\{\sigma_i(M)\}_i$ are the singular values of M , i.e., the eigenvalues of the positive semi-definite operator $|M| \triangleq \sqrt{M^\dagger M}$. Moreover, let $\|M\| \triangleq$

Figure 1: Future work item 4 in arXiv:2109.02600, PDF page 9.

This packet proves such a physical-space theorem for all finite exponents $1 \leq p \leq q < \infty$, with explicit dimension-free constants. No optimality claim is made for the constants outside the endpoints inherited from known theorems.

Notation

Let $G = \mathbb{Z}_r^n$ with normalized counting measure. Put $\omega_r = e^{2\pi i/r}$ and write

$$\chi_S(x) = \omega_r^{S \cdot x}, \quad S \cdot x = \sum_{j=1}^n S_j x_j \pmod{r}.$$

For $f : G \rightarrow \mathbb{C}^{m \times m}$ define

$$\widehat{f}(S) = \mathbb{E}_{x \in G} f(x) \overline{\chi_S(x)}, \quad T_\rho f = \sum_{S \in G} \rho^{|S|} \widehat{f}(S) \chi_S,$$

where $|S| = |\{j : S_j \neq 0\}|$. For $1 \leq s < \infty$ let

$$\|f\|_{L_s(S_s)} = (\mathbb{E}_{x \in G} \|f(x)\|_s^s)^{1/s},$$

where $\|\cdot\|_s$ denotes the unnormalized Schatten s -norm.

For $1 \leq s \leq 2$ set

$$\zeta_r(s) = \frac{(s-1)(1-(s-1)^{r-1})}{(r-1)(2-s)}$$

with the continuous values $\zeta_r(1) = 0$ and $\zeta_r(2) = 1$, and put

$$\beta_r(s) = \sqrt{\zeta_r(s)}.$$

Equivalently, for $t = s - 1 \in [0, 1]$,

$$\zeta_r(s) = \frac{t(1+t+\dots+t^{r-2})}{r-1},$$

so $0 \leq \beta_r(s) \leq 1$.

Admissible Constants

For $1 \leq p \leq q < \infty$ define $c_r(p, q)$ as follows.

If $p = q$, let $c_r(p, q) = 1$.

If $1 \leq p < q \leq 2$, set

$$\theta_q = 2 \left(1 - \frac{1}{q}\right), \quad \frac{1}{a(p, q)} = \frac{\frac{1}{p} + 1 - \frac{2}{q}}{2 - \frac{2}{q}},$$

and let

$$c_r(p, q) = \beta_r(a(p, q)).$$

Here $a(p, q) \in [1, 2]$; indeed $a(1, q) = 1$ and $a(q, q) = 2$.

If $1 \leq p \leq 2 \leq q < \infty$, the preceding definitions agree on the boundary and we set

$$c_r(p, q) = \beta_r(p) \beta_r(q'), \quad q' = \frac{q}{q-1}.$$

If $2 \leq p < q < \infty$, set

$$c_r(p, q) = c_r(q', p'), \quad p' = \frac{p}{p-1}, \quad q' = \frac{q}{q-1}.$$

Proof Intuition

The source theorem is stated in a Fourier-coefficient form for $1 \leq s \leq 2$. Parseval converts its target side into a physical $L_2(S_2)$ estimate: the Hilbert-Schmidt norm is the coefficient norm for which Parseval is exact, and $\|A\|_2 \leq \|A\|_s$ for $s \leq 2$. Thus the source theorem already gives $L_s(S_s) \rightarrow L_2(S_2)$ hypercontractivity with noise $\beta_r(s)$.

The noise operator is also an $L_t(S_t)$ contraction for every $t \geq 1$, because it is convolution by a probability kernel. Interpolating between $L_1(S_1) \rightarrow L_1(S_1)$ contraction and $L_a(S_a) \rightarrow L_2(S_2)$ gives the whole range $1 \leq p \leq q \leq 2$. Duality gives $2 \leq p \leq q < \infty$. Finally, when $p \leq 2 \leq q$, compose the $p \rightarrow 2$ estimate with the dual $2 \rightarrow q$ estimate, using $T_{\rho_1} T_{\rho_2} = T_{\rho_1 \rho_2}$.

Endpoint Tools

Lemma 1 (Source coefficient theorem). *Let $1 \leq s \leq 2$. If $0 \leq \rho \leq \beta_r(s)$, then every $f : \mathbb{Z}_r^n \rightarrow \mathbb{C}^{m \times m}$ satisfies*

$$\left(\sum_{S \in \mathbb{Z}_r^n} \rho^{2|S|} \|\widehat{f}(S)\|_s^2 \right)^{1/2} \leq \|f\|_{L_s(S_s)}.$$

Proof. This is Result 1, equivalently Theorem 3.3, of Arunachalam and Doriguello with the notation above. The endpoint $s = 2$ is Parseval, and $s = 1$ follows from the displayed inequality by continuity or directly from convexity of the trace norm applied to the constant Fourier coefficient. $\square$

Lemma 2 (Physical $s \rightarrow 2$ endpoint). *Let $1 \leq s \leq 2$. If $0 \leq \rho \leq \beta_r(s)$, then*

$$\|T_\rho f\|_{L_2(S_2)} \leq \|f\|_{L_s(S_s)}$$

for every $f : \mathbb{Z}_r^n \rightarrow \mathbb{C}^{m \times m}$.

Proof. By Parseval in the Hilbert-Schmidt norm,

$$\|T_\rho f\|_{L_2(S_2)}^2 = \sum_{S \in \mathbb{Z}_r^n} \rho^{2|S|} \|\widehat{f}(S)\|_2^2.$$

Since $s \leq 2$, Schatten norm monotonicity gives $\|A\|_2 \leq \|A\|_s$ for every matrix A . Hence

$$\|T_\rho f\|_{L_2(S_2)}^2 \leq \sum_{S \in \mathbb{Z}_r^n} \rho^{2|S|} \|\widehat{f}(S)\|_s^2 \leq \|f\|_{L_s(S_s)}^2$$

by Lemma 1. $\square$

Lemma 3 (Noise contractions). *For every $1 \leq t < \infty$ and $0 \leq \rho \leq 1$,*

$$\|T_\rho f\|_{L_t(S_t)} \leq \|f\|_{L_t(S_t)}.$$

Proof. On one copy of $\mathbb{Z}_r$, the multiplier that is 1 on the trivial character and ρ on all nontrivial characters is convolution by the probability measure

$$\mu_\rho(0) = \frac{1 + (r-1)\rho}{r}, \quad \mu_\rho(y) = \frac{1-\rho}{r} \quad (y \neq 0).$$

On $\mathbb{Z}_r^n$ the kernel is the product measure $\mu_\rho^{\otimes n}$. Thus

$$T_\rho f(x) = \mathbb{E}_y f(x + y).$$

By convexity of $A \mapsto \|A\|_t^t$,

$$\|T_\rho f\|_{L_t(S_t)}^t = \mathbb{E}_x \|\mathbb{E}_y f(x + y)\|_t^t \leq \mathbb{E}_{x,y} \|f(x + y)\|_t^t = \|f\|_{L_t(S_t)}^t.$$

$\square$

Lemma 4 (Interpolation framework). *Suppose a linear map T is a contraction*

$$L_{u_0}(S_{u_0}) \rightarrow L_{v_0}(S_{v_0}) \quad \text{and} \quad L_{u_1}(S_{u_1}) \rightarrow L_{v_1}(S_{v_1})$$

for exponents in $[1, \infty)$. If $0 \leq \theta \leq 1$ and

$$\frac{1}{u} = \frac{1-\theta}{u_0} + \frac{\theta}{u_1}, \quad \frac{1}{v} = \frac{1-\theta}{v_0} + \frac{\theta}{v_1},$$

then $T : L_u(S_u) \rightarrow L_v(S_v)$ is also a contraction.

Proof. Identify functions $f : G \rightarrow \mathbb{C}^{m \times m}$ with the finite von Neumann algebra $\ell_\infty(G) \otimes M_m(\mathbb{C})$ equipped with the trace

$$\tau(f) = \mathbb{E}_{x \in G} \text{Tr } f(x).$$

Then $\|f\|_{L_s(S_s)} = \tau(|f|^s)^{1/s}$. The finite-dimensional noncommutative L_s spaces form the exact complex interpolation scale, so the claim is the usual Riesz-Thorin/complex interpolation theorem for these finite-dimensional noncommutative L_s spaces. $\square$

Main Theorem

Theorem 1 (Finite-exponent matrix (q, p) hypercontractivity). *Let $r \geq 2$, $n, m \geq 1$, and $1 \leq p \leq q < \infty$. If $0 \leq \rho \leq c_r(p, q)$, then every $f : \mathbb{Z}_r^n \rightarrow \mathbb{C}^{m \times m}$ satisfies*

$$\|T_\rho f\|_{L_q(S_q)} \leq \|f\|_{L_p(S_p)}.$$

Proof. We split the proof into three ranges.

The range $1 \leq p \leq q \leq 2$. If $p = q$, this is Lemma 3. Assume $p < q$. Put

$$\theta = \theta_q = 2 \left(1 - \frac{1}{q}\right), \quad a = a(p, q).$$

By definition,

$$\frac{1}{q} = (1 - \theta) \cdot 1 + \theta \cdot \frac{1}{2}, \quad \frac{1}{p} = (1 - \theta) \cdot 1 + \theta \cdot \frac{1}{a}.$$

If $\rho \leq c_r(p, q) = \beta_r(a)$, then T_ρ is an $L_1(S_1) \rightarrow L_1(S_1)$ contraction by Lemma 3, and an $L_a(S_a) \rightarrow L_2(S_2)$ contraction by Lemma 2. Interpolating with Lemma 4 gives the desired $L_p(S_p) \rightarrow L_q(S_q)$ contraction.

The range $2 \leq p \leq q < \infty$. Again $p = q$ is Lemma 3, so assume $p < q$. Then $1 < q' \leq p' \leq 2$. By the first part, if $\rho \leq c_r(q', p') = c_r(p, q)$, then

$$\|T_\rho g\|_{L_{p'}(S_{p'})} \leq \|g\|_{L_{q'}(S_{q'})}.$$

The operator T_ρ is self-adjoint for the trace pairing

$$\langle f, g \rangle = \mathbb{E}_{x \in G} \text{Tr}(f(x)^* g(x)),$$

because it is a real Fourier multiplier. Therefore finite-dimensional duality yields

$$\begin{aligned} \|T_\rho f\|_{L_q(S_q)} &= \sup_{\|g\|_{L_{q'}(S_{q'})} \leq 1} |\langle T_\rho f, g \rangle| \\ &= \sup_{\|g\|_{L_{q'}(S_{q'})} \leq 1} |\langle f, T_\rho g \rangle| \leq \|f\|_{L_p(S_p)}. \end{aligned}$$

The crossing range $1 \leq p \leq 2 \leq q < \infty$. If $p = q = 2$, this has already been covered. If $p = 1 < q$, then $c_r(1, q) = 0$, and for $\rho = 0$ the operator T_0 is projection onto the constant function $\mathbb{E}f$. Jensen's inequality and $\|A\|_q \leq \|A\|_1$ give

$$\|T_0 f\|_{L_q(S_q)} = \|\mathbb{E}f\|_q \leq \mathbb{E}\|f\|_q \leq \mathbb{E}\|f\|_1 = \|f\|_{L_1(S_1)}.$$

Now assume $p > 1$. Let

$$b_p = \beta_r(p), \quad b_q = \beta_r(q').$$

If $\rho \leq b_p b_q$, choose $\rho_1 = b_p$ and $\rho_2 = \rho/b_p$; then $0 \leq \rho_2 \leq b_q$ and

$$T_\rho = T_{\rho_2} T_{\rho_1}.$$

By Lemma 2, T_{ρ_1} is a contraction $L_p(S_p) \rightarrow L_2(S_2)$. By the already proved dual estimate applied to the endpoint $q' \rightarrow 2$, T_{ρ_2} is a contraction $L_2(S_2) \rightarrow L_q(S_q)$. Hence

$$\|T_\rho f\|_{L_q(S_q)} \leq \|T_{\rho_1} f\|_{L_2(S_2)} \leq \|f\|_{L_p(S_p)}.$$

This proves all cases. □

Remark 1 (Boolean constants). *For $r = 2$ one has $\beta_2(s) = \sqrt{s-1}$. Thus in the crossing range $p \leq 2 \leq q$,*

$$c_2(p, q) = \sqrt{p-1} \sqrt{q'-1} = \sqrt{\frac{p-1}{q-1}},$$

the classical Nelson threshold. The interpolation and duality constants below and above 2 are dimension-free but are not claimed to be optimal.

Novelty Check and Limitations

The packet proves the finite-exponent physical-space version of the generalized matrix hypercontractivity question over $\mathbb{Z}_r^n$, and therefore also over the Boolean cube by taking $r = 2$. It does not assert an L_∞ endpoint, nor does it assert sharp constants for all finite pairs (p, q) .

The cheap run indexes were searched for arXiv:2109.02600, the title, matrix hypercontractivity, generalized (q, p) hypercontractivity, Schatten norms, and $\mathbb{Z}_r$. The only relevant hits were the two earlier lane 19 partial packets. A bounded web check on July 18, 2026 searched exact combinations of these same terms. I found the source paper and standard background, but no exact later finite-exponent physical-space theorem matching Theorem 1.

References

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